

Part III Quantum Field Theory

Introduction

This PDF is a transcription of my handwritten notes for the Part III Quantum Field Theory course offered during Michaelmas term 2025 at the University of Cambridge. This course was lectured by Alejandra Castro but any typos or mistakes in these notes are strictly *my fault!* These notes are organised lecture-wise.

Lecture 1

A major theme of 20th century physics was the construction of physical theories which are local. Locality emerges naturally in so-called *field theories*, of which Maxwell's electrodynamics and Einstein's theory of general relativity are classical examples. One motivation for quantum field theory (QFT) is to formulate quantum mechanics in a way which is compatible with notions of locality, as described by Einstein's relativity.

In this course, we work with the natural unit convention; space and time have equal dimension (meaning that speeds are dimensionless). In particular, the speed of light in vacuum is set to one $c = 1$. Also, we set the reduced Planck constant equal to one $\hbar = 1$. Setting $\hbar = 1$ in addition to $c = 1$ implies that energies have dimension equivalent to an inverse length (inverse time). As is standard in particle physics, we use a mostly-minus signature for the Minkowski metric, $\eta_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$.

review of classical field theory

It's well-known that classical mechanics can be formulated in terms of Hamilton's variational principle. If our system is described at time t by generalised coordinates $q = (q_1, q_2, \dots)$ and generalised speeds $\dot{q} = (\dot{q}_1, \dot{q}_2, \dots)$, we suppose the existence of a so-called *Lagrangian* function $L(q, \dot{q}; t)$ which encodes all dynamics. In particular, we define the *action* to be the temporal integral

$$S_{[q]} = \int_{t_i}^{t_f} dt L(q, \dot{q}; t),$$

where it is understood that the action S is a functional of the trajectory q . Hamilton's variational principle states that the physical trajectory $\tilde{q}$ extremises the action

$$\left. \frac{\delta S}{\delta q} \right|_{q=\tilde{q}} = 0.$$

The standard way to pose this problem is to say, given some Lagrangian $L(q, \dot{q}; t)$ and boundary conditions $q(t_i) = q_i$ and $q(t_f) = q_f$, how does one find the physical trajectory $\tilde{q}$? The answer to this question is to write $q = \tilde{q} + \delta q$ where δq is, by some metric, a small deformation about the extremal trajectory which respects boundary conditions ($\delta q(t) = 0$ at $t = t_i, t_f$). By expanding the Lagrangian to first order in δq we see

$$\delta S = \int_{t_i}^{t_f} dt \left\{ \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right\}.$$

Note that we've restricted our attention to Lagrangians which depend at most on *first* derivatives of the generalised coordinates. It's perfectly valid to study Lagrangians depending on higher-order derivatives of q but these yield equations of motion which are strictly more than second-order – something physicists traditionally avoid. There is an implied summation over all the coordinate indices (i.e. you should read $\partial L / \partial q \times \delta q$ as $\sum_i \partial L / \partial q_i \times \delta q_i$ and so on). Noting that $\delta \dot{q} = d\delta q / dt$, we apply the product rule to find

$$\delta S = \int_{t_i}^{t_f} dt \left\{ \frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right\} \delta q + \left. \frac{\partial L}{\partial \dot{q}} \delta q \right|_{t_i}^{t_f},$$

where the boundary term on the right vanishes by our assertion that $\delta q(t) = 0$ when $t = t_i, t_f$. Hamilton's principle is then satisfied iff the so-called Euler-Lagrange (EL) equations hold true

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0.$$

By substituting our particular Lagrangian into the EL equation and simplifying, we obtain the system's so-called *equations of motion*.

In the next lecture, we adapt this variational framework to a field degree-of-freedom (i.e. we treat the generalised coordinate as a function ϕ on Minkowski spacetime). This involves increasing the number of parameters integrated over in the action from one (just the time) to four (time + three spatial dimensions). Application of Hamilton's variational principle will then produce EL equations which constrain the spacetime evolution of the field.

Importantly, the symmetries of the system are reflected in the symmetries of the action. If some transformation of the system leaves the action unchanged up to some total derivative, the resulting EL equations (hence the equations of motion) remain invariant. This is what it means for a particular transformation to be a symmetry of our model. In quantum field theory, we assert that all Lagrangians must transform as scalars under Lorentz transformations (the group of rotations + boosts).

Lecture 2

To proceed, we need to construct an action principle for a field degree-of-freedom. In the previous lecture, we discussed a *one-dimensional* theory (there was only one parameter being integrated over in S , the time). The standard approach in field theory is to treat the spatial coordinates x^i as being parameters placed on equal footing with the time t . Indeed, we adopt the labelling convention $(t, x^1, x^2, x^3) \mapsto (x^0, x^1, x^2, x^3)$ for spacetime vectors. Therefore our theory is *four-dimensional*.

In this instance, the action is expressed as a spacetime integral

$$S = \int d^4x \mathcal{L}(\phi, \partial\phi),$$

where $\mathcal{L}$ is the so-called *Lagrangian density* (although most people still call it the Lagrangian) which is related to the Lagrangian via a spatial integral $L = \int d^3x \mathcal{L}$ but this requires a particular choice of frame and is, of course, not Lorentz invariant. The Lagrangian density is, by construction, a Lorentz scalar. The field ϕ appearing as an argument to $\mathcal{L}$ is, in the simplest case, a real-valued function on Minkowski space $\phi : \mathbb{M} \rightarrow \mathbb{R}$. We define the 4-gradient in the natural way $\partial_\mu = (\partial/\partial x^0, \partial/\partial x^1, \partial/\partial x^2, \partial/\partial x^3)$. One can find Euler-Lagrange equations for the four-dimensional theory using the same arguments as for the one-dimensional theory (indeed, the argument from lecture one generalises to arbitrary dimensions). The resulting Euler-Lagrange equation is

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right) = 0,$$

where this equation holds true for every field degree-of-freedom at play in $\mathcal{L}$ (if you like, we can think of ϕ as implicitly indexing multiple fields ϕ_a , each having its own EL equation).

As remarked, the Lagrangian density is designed to be a Lorentz scalar. This means that $\mathcal{L}(\phi(x), \partial\phi(x)) = \mathcal{L}(x)$ transforms under the trivial representation of the Lorentz group. Symbolically $\mathcal{L}(x) \mapsto \mathcal{L}(\Lambda^{-1}x)$ for Lorentz transformations Λ . When this is true, it's automatically the case that Lorentz transformations are a symmetry of the action (the Jacobian incurred by any Lorentz transformation is one).

The simplest example we can study is the real scalar field $\phi : \mathbb{M} \rightarrow \mathbb{R}$ with Lagrangian density

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2.$$

The prefactors of $1/2$ are conventional (the equations of motion are generally unchanged by a constant rescaling of $\mathcal{L}$). The particular form of $\mathcal{L}$ as a difference between derivative terms and a polynomial in ϕ has its roots in classical Lagrangian mechanics – I recommend the interested reader to consult Landau's book on classical mechanics. Substituting this form of $\mathcal{L}$ into the EL equation produces the following e.o.m.

$$(\square + m^2)\phi = 0,$$

where $\square = \partial_\mu \partial^\mu$. This wave equation is famously known as the *Klein–Gordon* equation. Note that at this stage there is nothing quantum going on – we are still firmly within the regime of classical field theory.

Another example is the Lagrangian for a free electromagnetic field described by gauge potential A_μ . In particular

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu},$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The unsourced Maxwell equations are then encoded by

$$\partial_\mu F^{\mu\nu} = 0 \qquad \partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0,$$

where the first equation is the e.o.m. following from EL and the second equation is a Bianchi identity.

The Hamiltonian density $\mathcal{H}$ associated to $\mathcal{L}$ is found via the standard Legendre transform. Conjugate to any field d.o.f. ϕ_a is a canonical momentum Π_a which is defined

$$\Pi_a = \frac{\partial \mathcal{L}}{\partial \partial_0 \phi_a}.$$

The Hamiltonian density for the field is then

$$\mathcal{H} = \sum_a (\partial_0 \phi_a) \Pi_a - \mathcal{L},$$

which is familiar from a first course on classical mechanics. As usual, we prefer to speak about the canonical pair ϕ_a and Π_a when describing the field dynamics as opposed to ϕ_a and $\partial_0 \phi_a$ (although it's certainly possible that these coincide in particular examples). Just as the full Lagrangian L is related to the Lagrangian density $\mathcal{L}$ via a spatial integral, so is the full Hamiltonian H related to the Hamiltonian density $\mathcal{H}$ via a spatial integral $H = \int d^3x \mathcal{H}$. For the massive real scalar field discussed earlier, it is straightforward to find

$$H = \frac{1}{2} \int d^3x \{ \Pi^2 + (\nabla \phi)^2 + m^2 \phi^2 \}.$$

Lorentz symmetry

Underpinning all our QFTs is the assumption that they respect Lorentz symmetry (in practical terms, this means that the e.o.m. are unchanged by Lorentz transformations). The defining equation for the group of Lorentz transformations is that any Lorentz transformation Λ satisfies

$$\Lambda^T \eta \Lambda = \eta,$$

with η of course being the metric. It is easy to see that the set of Lorentz transformations form a (Lie) group. Moreover, we can take the determinant of the above equation to find $\det \Lambda = \pm 1$ which decomposes

the Lorentz group into at least two disconnected components (recall that the determinant is a continuous function of the matrix entries, so must vary continuously over any component of a Lie group). Moreover, the invariance of η under Lorentz transformations implies $(\Lambda_{00})^2 \geq 1$ which splits the Lorentz group into two further disconnected pieces. In total, there are four disconnected components which make up the Lorentz group, with the component containing the identity being the only subgroup. We call the component containing the identity the proper orthochronous Lorentz group (proper because $\det$ is positive here – handedness is not changed – and orthochronous because $\Lambda_{00} \geq 1$; time direction is not flipped).

We can make more statements about the Lorentz group by studying its algebra. Supposing that Λ is, by some standard, ‘close’ to the identity we can expand

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \omega^\mu_\nu + O(\omega^2),$$

where ω is the small deformation about the identity transformation and is, in general, some linear combination of generators of the Lorentz algebra. By supposition, Λ is a Lorentz transformation so

$$\begin{aligned}\eta_{\mu\nu} &= \Lambda^\rho_\mu \eta_{\rho\tau} \Lambda^\tau_\nu \\ &= (\delta^\rho_\mu + \omega^\rho_\mu) \eta_{\rho\tau} (\delta^\tau_\nu + \omega^\tau_\nu) + O(\omega^2) \\ &= \eta_{\mu\nu} + \omega_{\mu\nu} + \omega_{\nu\mu} + O(\omega^2),\end{aligned}$$

which, to hold at $O(\omega)$, requires that $\omega_{\mu\nu} = -\omega_{\nu\mu}$. Equivalently, the generators of the Lorentz algebra must be antisymmetric. In a four-dimensional spacetime, this implies that there are $4(4-1)/2 = 6$ degrees of freedom available to us when specifying a Lorentz transformation. If we like, we can split these into three independent spatial rotations and three independent boosts.

Generically, the fields we use to construct our Lagrangian transform under a representation D of the Lorentz group; $\phi_a(x) \mapsto D[\Lambda]_a^b \phi_b(\Lambda^{-1}x)$. The scalar field we discussed earlier transforms under the trivial representation of the Lorentz group ($D[\Lambda] = 1$ for all Λ). Valid Lagrangians are those that transform trivially under Lorentz transformations (up to the addition of a total derivative).

Lecture 3

A foundational result in Lagrangian mechanics is Noether’s theorem; every continuous symmetry of the action gives rise to a conserved current J^μ (conserved in the sense $\partial_\mu J^\mu = 0$). Every conserved current is associated to a conserved charge $Q = \int d^3x J^0$ provided the flux of J^i vanishes across the boundary of the system.

A transformation $\phi_a \mapsto \phi_a + \delta\phi_a$ is considered a symmetry of the action if and only if $S[\phi'] - S[\phi] = 0$. Generally, this allows the Lagrangian to change only by a total derivative on spacetime; $\mathcal{L} \mapsto \mathcal{L} + \partial_\mu F^\mu$. The term F^μ vanishes with $\delta\phi$ which, by assumption, is zero on the boundary. A simple application of the chain rule tells us

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi} \delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi} \delta\partial_\mu\phi,$$

a further application of the product rule tells us

$$\delta\mathcal{L} = \left(\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \left(\frac{\mathcal{L}}{\partial\partial_\mu\phi} \right) \right) \delta\phi + \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi} \delta\phi \right).$$

The term proportional to $\delta\phi$ vanishes on-shell (i.e. when the e.o.m. are satisfied) which implies that $\delta\mathcal{L} = \partial_\mu (\partial\mathcal{L}/\partial\partial_\mu\phi \times \delta\phi)$, which necessarily has the structure of a total derivative. Supposing that substitution of the transformed field $\phi + \delta\phi$ into $\mathcal{L}$ produces $\mathcal{L} + \partial_\mu F^\mu$, we can identify the two different derivatives to find

$$\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi} \delta\phi - F_\mu \right) = 0,$$

which allows us to identify the parenthetical expression as a conserved current J^μ . If J^μ is contracted against constants we can strip these off to form higher-rank conserved currents (e.g. the energy-momentum tensor $T^{\mu\nu}$). Next, we can show that the conserved charge Q is time-independent

$$\begin{aligned}\frac{dQ}{dt} &= \frac{d}{dt} \int d^3x J^0 \\ &= \int d^3x \frac{\partial J^0}{\partial t} \\ &= - \int d^3x \nabla_i J^i \\ &= - \int_{\text{boundary}} dA_i J^i,\end{aligned}$$

which follows from $\partial_\mu J^\mu = \partial_t J^0 + \nabla_i J^i = 0$ and the divergence theorem. The current ordinarily vanishes with $\delta\phi$, which is always chosen zero on the system's spacetime boundary. $dQ/dt = 0$ follows.

example: the energy-momentum tensor

The energy-momentum tensor arises as the conserved current associated to global spacetime translations; $x \mapsto x - \epsilon$ for small displacements ϵ . Taylor expanding $\phi(x + \epsilon)$ to $\mathcal{O}(\epsilon)$ gives

$$\phi(x + \epsilon) = \phi(x) + \epsilon^\mu \partial_\mu \phi(x) + \mathcal{O}(\epsilon^2),$$

with a similar expansion for the Lagrangian

$$\mathcal{L}(x + \epsilon) = \mathcal{L}(x) + \epsilon^\mu \partial_\mu \mathcal{L}(x) + \mathcal{O}(\epsilon^2).$$

From the expansion of the field and Lagrangian, we can identify $\delta\phi = \epsilon^\mu \partial_\mu \phi$ and $\delta\mathcal{L} = \partial_\mu(\epsilon^\mu \mathcal{L})$. Using our result from the previous lecture, we find the conserved current as

$$\begin{aligned}J^\mu &= \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta\phi - \epsilon^\mu \mathcal{L} \\ &= \epsilon^\nu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - \delta_\nu^\mu \mathcal{L} \right).\end{aligned}$$

Stripping off the constant components ϵ^ν and raising the ν index, we find the energy-momentum tensor as

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L},$$

where you should keep in mind that $T^{\mu\nu}$ remains a function on spacetime, $\mathbb{M}$. The equation $\partial_\mu T^{\mu\nu} = 0$ explicates the existence of four conserved currents (corresponding to the four choices of the free index ν).